

Distributivity and the Commensurability of Value-Definite Realism and Empirical Adequacy

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Abstract

When does an empirically adequate theory admit a value-definite completion — an assignment of simultaneous definite values to all observables? We show that the answer depends on the algebraic structure of the observations. For a directed system of Boolean algebras carrying compatible charges, the Stone construction produces an unconditional measure on the dual space, and every observable admits a simultaneous definite value via ultrafilters. The passage to a probability measure on an underlying state space requires only σ -additivity, and the choice of state space is unconstrained by the algebra [9].

For orthomodular lattices — the algebraic framework for incompatible observations — the situation is structurally different. The McDonald–Bimbó dual space carries principal filters as distinguished structure, and the Kochen–Specker theorem blocks global value-definiteness. Probabilistic realism (descent to a probability measure) remains available via Gleason-type results, but value-definite realism does not.

The dividing line is distributivity. We identify a four-level coherence hierarchy — consistency, contextual coherence, probabilistic coherence, value-definite coherence — and show that distributivity of the observation algebra is the exact condition under which all four levels are mutually commensurable.

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1 Introduction

A central question in the philosophy of science asks when an empirically successful theory can be completed to a realist one — one that assigns definite properties to an underlying reality, not merely adequate predictions to observable phenomena. Van Fraassen [11] argues that science need only aim at *empirical adequacy*: agreement with all observable phenomena. The scientific realist demands more — an underlying state space whose properties generate the observations.

This debate is usually conducted in philosophical terms. We give it a mathematical formulation and show that its resolution depends on the algebraic structure of the observations.

Three kinds of realism

We distinguish three nested positions, each a strengthening of the previous:

- (i) **Empirical adequacy (EA).** A measure on the dual space of the observation algebra that agrees with all observed charges. No commitment to an underlying state space.
- (ii) **Probabilistic realism (PR).** A probability measure on an underlying state space $(\Omega, \sigma(\mathcal{C}))$ that generates the observations. Requires descent from the dual-space measure.

- (iii) **Value-definite realism (VDR).** Every observable simultaneously has a definite value. A global dispersion-free state — equivalently, a 2-valued homomorphism on the full observation algebra.

These are nested: VDR implies PR implies EA. The question is which implications reverse, and what governs the answer.

The role of the observation algebra

Consider two settings:

Compatible observations. When all observations can be performed simultaneously, the observation algebra is a Boolean algebra — a complemented distributive lattice. Every Boolean algebra admits 2-valued homomorphisms (ultrafilters exist by Zorn’s lemma). In this setting, all three positions are available and mutually commensurable. The passage from EA to PR requires σ -additivity of the charges [9]; the further passage to VDR is free.

Incompatible observations. When some observations cannot be performed simultaneously — as in quantum mechanics, contextual experimental design, or distributed systems with interfering sensors — the observation algebra is an orthomodular lattice (OML), a complemented but non-distributive lattice. The Kochen–Specker theorem [8] shows that for the prototypical case $L(H)$ with $\dim H \geq 3$, no 2-valued homomorphism exists: VDR is blocked. Probabilistic realism remains available via Gleason’s theorem [5], and empirical adequacy is unconditional.

The dividing line is distributivity.

Contributions

We develop the following:

1. A unified framework based on directed systems of complemented lattices, with the Boolean and orthomodular cases as instances (§2–§4).
2. A four-level coherence hierarchy — consistency, contextual coherence, probabilistic coherence, value-definite coherence — with the obstruction at each level identified (§8).
3. A structural theorem: distributivity of the observation algebra is the exact condition under which all four levels are mutually commensurable (§7).
4. An analysis of the philosophical consequences: the realism/empiricism debate has different mathematical answers depending on the observation algebra (§9).

2 The Boolean case

We recall the framework and results of [9], which treats directed systems of Boolean algebras.

Directed systems and compatible charges

Let $(\iota, \leq)$ be a directed set indexing a family of Boolean algebras $\{B_i\}_{i \in \iota}$ with injective homomorphisms $\varphi_{ij} : B_i \rightarrow B_j$ for $i \leq j$. Write $\mathcal{C} = \bigcup_i B_i$ for the direct limit — the *cylinder algebra*.

A *compatible family of normalised charges* is a collection $\{\ell_i\}$ with $\ell_i : B_i \rightarrow [0, 1]$ finitely additive, $\ell_i(1) = 1$, and $\ell_j \circ \varphi_{ij} = \ell_i$ for $i \leq j$.

The Stone construction

The Stone space $\text{St}(\mathcal{C})$ is the space of ultrafilters on $\mathcal{C}$, compact and totally disconnected. Each $\omega \in \Omega$ determines a principal ultrafilter $\text{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$.

Any compatible family of normalised charges extends to a σ -additive Baire probability measure $\hat{\mu}$ on $\text{St}(\mathcal{C})$ — unconditionally, with no assumption of σ -additivity [3, 9]. This is the *empirically adequate* object: it agrees with all charges on cylinder events.

Descent and σ -additivity

The measure $\hat{\mu}$ lives on $\text{St}(\mathcal{C})$, not on Ω . Descent to a probability measure P on $(\Omega, \sigma(\mathcal{C}))$ requires $\hat{\mu}(\text{pure}(\Omega)) = 1$ — that $\hat{\mu}$ concentrates on the principal ultrafilters. This holds if and only if each charge ℓ_i is σ -additive [9].

Value-definiteness in the Boolean case

Every Boolean algebra admits ultrafilters (by Zorn's lemma), and every ultrafilter is a 2-valued homomorphism. A 2-valued homomorphism assigns to each element $a \in \mathcal{C}$ a definite value in $\{0, 1\}$, respecting the lattice operations.

Moreover, any subset $S \subseteq \text{St}(\mathcal{C})$ that projects surjectively onto each $\text{St}(B_i)$ can serve as a realisation space: set $\Omega = S$ with evaluation maps given by the restriction maps. The algebra imposes no constraint on the choice of Ω [9].

Proposition 2.1 (Boolean commensurability). *For a directed system of Boolean algebras with compatible charges:*

- (a) *EA is unconditional: $\hat{\mu}$ on $\text{St}(\mathcal{C})$ always exists.*
- (b) *PR holds iff each ℓ_i is σ -additive.*
- (c) *VDR holds unconditionally: $\text{St}(\mathcal{C})$ consists entirely of 2-valued homomorphisms.*

In particular, EA, PR, and VDR are mutually commensurable: the only obstruction is σ -additivity at the EA–PR transition, and it is a condition on the charges, not the algebra.

Proof. Part (a) is the Stone construction [3, 9]. Part (b) is [9, Theorem 1]. Part (c): ultrafilters on a Boolean algebra are exactly the 2-valued homomorphisms; their existence follows from Zorn's lemma applied to the filter generated by any non-zero element. $\square$

3 Orthomodular observation algebras

The Boolean assumption is that all observations are compatible: any finite collection can be performed simultaneously. When this fails — when some observations preclude others — the observation algebra loses distributivity. The appropriate generalisation is the orthomodular lattice.

Orthomodular lattices

Definition 3.1. An *ortholattice* is a bounded lattice $A = (A, \wedge, \vee, {}^\perp, 0, 1)$ with an orthocomplementation ${}^\perp : A \rightarrow A$ satisfying $a \wedge a^\perp = 0$, $a \vee a^\perp = 1$, $a^{\perp\perp} = a$, and $a \leq b \Rightarrow b^\perp \leq a^\perp$.

Definition 3.2. An *orthomodular lattice* (OML) is an ortholattice satisfying the *orthomodular law*:

$$a \leq b \implies b = a \vee (a^\perp \wedge b).$$

Every Boolean algebra is an OML (the orthomodular law follows from distributivity), but the converse fails. An OML is Boolean if and only if it is distributive [10, Proposition 2.3].

Example 3.3. Let H be a separable Hilbert space with $\dim H \geq 3$. The lattice $L(H)$ of closed linear subspaces, ordered by inclusion with $U^\perp = \{x \in H : \langle x, y \rangle = 0 \ \forall y \in U\}$, is an orthomodular lattice that is not distributive [7].

Example 3.4 (Incompatible experimental design). Consider an experiment with three binary measurements A, B, C , where (A, B) and (A, C) can be performed jointly but (B, C) cannot. The jointly measurable pairs generate Boolean subalgebras, but the full observation algebra is orthomodular, not Boolean. This situation arises in contextual behavioural science, clinical trials with mutually exclusive protocols, and sensor networks with interfering instruments.

Commutativity and classical contexts

Two elements a, b in an OML are said to *commute* if they generate a Boolean subalgebra. A maximal set of pairwise commuting elements forms a *Boolean context* — a maximal Boolean subalgebra of A . Within each context, classical logic and classical probability apply. The non-classical character of the OML arises from the relationships *between* contexts.

States on OMLs

Definition 3.5. A *state* on an OML A is a function $s : A \rightarrow [0, 1]$ with $s(1) = 1$ and

$$a \perp b \implies s(a \vee b) = s(a) + s(b),$$

where $a \perp b$ means $a \leq b^\perp$.

A state is additive on *orthogonal* pairs only — not on arbitrary joins. This is the OML analogue of a finitely additive charge on a Boolean algebra.

Definition 3.6. A state s is *dispersion-free* if $s(a) \in \{0, 1\}$ for all $a \in A$.

A dispersion-free state is a 2-valued homomorphism: it assigns a definite truth value to every proposition in A . On a Boolean algebra, dispersion-free states are exactly the ultrafilters.

The Kochen–Specker obstruction

Theorem 3.7 (Kochen–Specker [8]). *If $\dim H \geq 3$, there is no dispersion-free state on $L(H)$.*

No assignment of simultaneous definite values to all observables is consistent with the algebraic structure. This is not a limitation of knowledge but a structural impossibility: the OML $L(H)$ does not admit a 2-valued homomorphism.

The obstruction is dimensional: for $\dim H = 2$, dispersion-free states exist (every maximal Boolean subalgebra of $L(\mathbb{C}^2)$ is a four-element Boolean algebra, and 2-valued homomorphisms are plentiful). The critical threshold is $\dim H = 3$.

Gleason’s theorem

Despite the absence of dispersion-free states, *probabilistic* states on $L(H)$ are well-behaved:

Theorem 3.8 (Gleason [5]). *If $\dim H \geq 3$, every σ -additive state on $L(H)$ is of the form $s(P) = \text{tr}(\rho P)$ for a unique density operator ρ on H .*

Gleason’s theorem says that probabilistic realism is available for $L(H)$: every state extends to a well-defined probability assignment via the Born rule. What fails is value-definite realism — the demand that every observable have a simultaneous definite value.

4 The OML dual space

In the Boolean case, Stone duality provides the dual space $\text{St}(\mathcal{C})$ of ultrafilters. For OMLs, the analogue is the McDonald–Bimbó duality [10], which we now describe.

Filters on an OML

Let A be an OML. A *filter* on A is a non-empty subset $x \subseteq A$ that is upward closed ($a \in x$ and $a \leq b$ implies $b \in x$) and closed under finite meets ($a, b \in x$ implies $a \wedge b \in x$). A filter is *proper* if $0 \notin x$; the unique *improper* filter is $\omega = A$.

For each $a \in A$, the *principal filter* generated by a is $\uparrow a = \{b \in A : a \leq b\}$. Write $F(A)$ for the collection of all filters on A , and $P(A) \subseteq F(A)$ for the subcollection of principal filters.

The dual space

Definition 4.1 (McDonald–Bimbó [10]). Let A be an OML. The *dual space* of A is the ordered relational topological space

$$S_0(A) = (F(A), \subseteq, \perp_A, P(A), \mathcal{T}),$$

where:

- (i) $F(A)$ is the set of all filters on A ;
- (ii) $\subseteq$ is set inclusion;
- (iii) $x \perp_A y$ iff there exists $a \in A$ with $a \in x$ and $a^\perp \in y$;
- (iv) $P(A)$ is the collection of principal filters;
- (v) $\mathcal{T}$ is the topology generated by the subbasis $\{h(a) : a \in A\} \cup \{F(A) \setminus h(a) : a \in A\}$, where $h(a) = \{x \in F(A) : a \in x\}$.

Theorem 4.2 (McDonald–Bimbó [10]). $S_0(A)$ is an orthomodular space (compact, totally disconnected). The category of OMLs and homomorphisms is dually equivalent to the category of orthomodular spaces and continuous weak p -morphisms. Moreover, $A \cong \text{CO}(S_0(A))^\dagger$, the OML of clopen $\perp$ -stable subsets of $S_0(A)$.

Structural comparison with Stone duality

The critical differences are collected in the following table:

Feature	Boolean (Stone)	OML (McDonald–Bimbó)
Dual points	Ultrafilters	All filters
Distinguished subset	None	$P(A)$ (principal filters)
Clopens	Boolean algebra	OML
Orthogonality	Trivial	Non-trivial ($\perp_A$)
2-valued homomorphisms	Plentiful (Zorn)	May not exist (KS)

In Stone duality, the principal filters (i.e., the principal ultrafilters $\text{pure}(\omega)$ for $\omega \in \Omega$) are *invisible*: they play no distinguished role in the dual space. Which ultrafilters are principal depends on the choice of Ω [9].

In the McDonald–Bimbó duality, $P(A)$ is *part of the structure* of the dual space: the orthomodular frame conditions (Definition 4.1) explicitly reference $P(A)$. This is the algebraic formalisation of the observation that, for non-distributive algebras, the distinction between realised and unrealised observation-patterns is not freely chosen but is constrained by the algebra.

5 Three kinds of realism, revisited

We now have the framework to state the three positions precisely in both settings.

Definition 5.1. Let A be a complemented lattice (Boolean or orthomodular) with a state $s : A \rightarrow [0, 1]$.

- (i) **Empirical adequacy (EA):** s defines a consistent assignment $\mu(h(a)) = s(a)$ on the clopens of the dual space.
- (ii) **Probabilistic realism (PR):** μ extends to a σ -additive probability measure on the Baire σ -algebra of the dual space, concentrating on $P(A)$.
- (iii) **Value-definite realism (VDR):** There exists a dispersion-free state on A — a 2-valued homomorphism assigning simultaneous definite values to all elements.

Proposition 5.2 (OML status). *For $A = L(H)$ with $\dim H \geq 3$:*

- (a) *EA holds: every state defines a consistent assignment on $\text{CO}(S_0(A))^\dagger$.*
- (b) *PR holds: Gleason's theorem (3.8) gives a σ -additive measure via $s(P) = \text{tr}(\rho P)$.*
- (c) *VDR fails: the Kochen–Specker theorem (3.7) shows no dispersion-free state exists.*

Comparing with Proposition 2.1:

Position	Boolean	OML ($\dim \geq 3$)
EA	✓ (unconditional)	✓ (unconditional)
PR	✓ (iff σ -additive)	✓ (Gleason)
VDR	✓ (ultrafilters)	× (Kochen–Specker)

The single structural difference — distributivity versus non-distributivity — controls exactly the VDR row.

6 The descent problem for OMLs

In the Boolean case, the descent from $\hat{\mu}$ on $\text{St}(\mathcal{C})$ to P on $(\Omega, \sigma(\mathcal{C}))$ requires σ -additivity of the charges and a choice of realisation space Ω . The algebra imposes no constraint on Ω .

For OMLs, the descent problem has a different character. A state s on A defines $\mu(h(a)) = s(a)$ on the clopen $\perp$ -stable sets $\text{CO}(S_0(A))^\dagger$.

The extension obstacle

In the Boolean case, $\text{Clop}(\text{St}(\mathcal{C}))$ is a Boolean algebra of sets, and s is finitely additive on it. The Carathéodory–Choksi extension machinery applies: compactness of $\text{St}(\mathcal{C})$ gives σ -subadditivity, and the extension to a Baire measure is automatic.

In the OML case, $\text{CO}(S_0(A))^\dagger$ is an OML of sets — closed under $\cap$ and $^\perp$ but *not* under arbitrary $\cup$. The state s is additive on orthogonal pairs only. Carathéodory does not apply: there is no Boolean algebra of sets on which s is finitely additive.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason's theorem (Theorem 3.8) provides the extension: every state arises from a density operator, giving a σ -additive measure via the Born rule. For general OMLs, the extension problem is open.

Partial value-definite descent

Even when probabilistic descent succeeds (Gleason), value-definite descent does not. The Bub–Clifton theorem [2] addresses what *can* be salvaged:

Theorem 6.1 (Bub–Clifton [2]). *Given a state ρ on $L(H)$ and a preferred observable R , there is a unique maximal sublattice $D(\rho, R) \subseteq L(H)$ such that:*

- (i) $D(\rho, R)$ is a Boolean algebra;
- (ii) $D(\rho, R)$ contains the spectral projections of R ;
- (iii) The restriction of ρ to $D(\rho, R)$ is a mixture of dispersion-free states.

In our terminology: $D(\rho, R)$ is the maximal Boolean subalgebra on which VDR is available — the largest “classical context” in which all observables simultaneously have definite values, given the state ρ and the preferred observable R .

This is a *partial* and *context-dependent* descent. It is partial because $D(\rho, R) \subsetneq L(H)$: not all observables receive definite values. It is context-dependent because $D(\rho, R)$ depends on the choice of R — different preferred observables yield different maximal Boolean subalgebras.

The Bub–Clifton result thus sharpens the Kochen–Specker obstruction: VDR fails globally, but it succeeds locally — within each Boolean context. The failure is in the *global* assembly, not in any individual context.

7 Distributivity as the commensurability condition

We can now state the main structural result.

Theorem 7.1 (Commensurability). *Let A be a complemented lattice.*

(a) *If A is distributive (Boolean), then EA, PR, and VDR are mutually commensurable:*

- EA is unconditional (Stone construction).
- PR holds iff the charges are σ -additive [9].
- VDR holds unconditionally (ultrafilters exist).

Moreover, the choice of realisation space Ω is unconstrained by the algebra.

(b) *If A is non-distributive (OML with $A \cong L(H)$, $\dim H \geq 3$), then:*

- EA is unconditional.
- PR holds (Gleason, Theorem 3.8).
- VDR fails globally (Kochen–Specker, Theorem 3.7).
- VDR holds locally within each Boolean context (Bub–Clifton, Theorem 6.1).

Proof. Part (a) is Proposition 2.1. For part (b): EA follows from the existence of states on $L(H)$ and the McDonald–Bimbó dual [10]; PR from Gleason’s theorem; VDR-failure from Kochen–Specker; local VDR from Bub–Clifton. $\square$

Corollary 7.2. *Distributivity of the observation algebra is the exact condition under which value-definite realism and empirical adequacy are commensurable.*

Proof. If A is distributive, VDR holds (Theorem 7.1(a)). If A is non-distributive with $A \cong L(H)$ and $\dim H \geq 3$, VDR fails (Theorem 7.1(b)). Since an OML is Boolean iff it is distributive [10], the dividing line is exactly distributivity. $\square$

Remark 7.3. The corollary requires the dimensional condition $\dim H \geq 3$ for the Kochen–Specker direction. For $\dim H = 2$, the OML $L(\mathbb{C}^2)$ is non-distributive but admits dispersion-free states: VDR holds despite non-distributivity. The sharp characterisation is: VDR is commensurable with EA iff A is distributive *or* every non-distributive component has dimension at most 2. In physically relevant cases ($\dim H \geq 3$), distributivity is the effective dividing line.

8 The coherence hierarchy

The preceding analysis reveals a four-level hierarchy of coherence conditions on an observation algebra A , each strictly stronger than the last.

Definition 8.1 (Coherence levels). Let A be a complemented lattice with a family of charges.

- Level 1. Consistency.** No contradiction within any single Boolean context: each maximal Boolean subalgebra of A admits a 2-valued homomorphism compatible with the charges.
- Level 2. Contextual coherence.** The assignments across all Boolean contexts are compatible: a state $s : A \rightarrow [0, 1]$ exists, agreeing with the charges on each context.
- Level 3. Probabilistic coherence.** The state extends to a σ -additive probability measure on the dual space of A — i.e., PR holds.
- Level 4. Value-definite coherence.** A global dispersion-free state exists: a 2-valued homomorphism on the full algebra A — i.e., VDR holds.

Each level implies the previous one: $4 \Rightarrow 3 \Rightarrow 2 \Rightarrow 1$. The converses fail in general. The following table records which transitions are obstructed:

Transition	Boolean	OML ($\dim \geq 3$)	Obstruction
$1 \rightarrow 2$	Trivial	Non-trivial	Contextuality
$2 \rightarrow 3$	σ -additivity	Gleason-type	Infinitary
$3 \rightarrow 4$	Free (ultrafilters)	Blocked	Kochen–Specker

In the Boolean case, transitions $1 \rightarrow 2$ and $3 \rightarrow 4$ are trivial: distributivity collapses them. The only non-trivial transition is $2 \rightarrow 3$, which is the content of [9].

In the OML case, every transition is non-trivial. The critical obstruction is at $3 \rightarrow 4$: the Kochen–Specker theorem blocks the passage from probabilistic coherence to value-definite coherence. This is the precise level at which distributivity is load-bearing.

Remark 8.2 (Relation to contextuality hierarchies). Abramsky and Brandenburger [1] introduced a hierarchy of contextuality: probabilistic, possibilistic (logical), and strong contextuality. Their hierarchy grades the *strength of the obstruction* to a global section of a presheaf of probability distributions. Our hierarchy grades *coherence demands*: what kind of global object the observation algebra can support. The two hierarchies are complementary and orthogonal: theirs asks “how badly does a global section fail?”; ours asks “what kind of global object can the algebra support?”

9 Discussion

The realism debate is not uniform

The standard philosophical debate between scientific realism and constructive empiricism treats the question as uniform: either theories aim at truth or they aim at empirical adequacy. The algebraic analysis above shows that the question is *parametrised* by the observation algebra.

For Boolean observations, the debate is genuine: the mathematics does not prefer one position over another. EA, PR, and VDR are all available; σ -additivity is the only bridge condition, and it is a property of the charges, not of the algebra.

For non-Boolean observations, the mathematics is not neutral. VDR is algebraically impossible (for $\dim \geq 3$); PR is available but structurally constrained; EA is always available. The constructive empiricist position [11] is the generic one. Value-definite realism is the special case — the case in which the observation algebra happens to be distributive.

Historical echoes

The distinction among the three positions illuminates classical debates:

- *Einstein’s realism* (“God does not play dice”) is the demand for VDR. The Kochen–Specker theorem shows this is not merely philosophically contestable but algebraically impossible for $\dim \geq 3$.
- *Bohr’s complementarity* is the observation that different Boolean contexts cannot be simultaneously realised. In our hierarchy, this is the non-triviality of the $1 \rightarrow 2$ transition.
- *Bell’s beables* [6] — the observables that *can* have definite values — are exactly the elements of the maximal Boolean subalgebra $D(\rho, R)$ identified by Bub–Clifton (Theorem 6.1).
- *De Finetti’s operationalism* is EA in the Boolean case: probability as coherent betting rates on observables, without commitment to an underlying state space.

Beyond quantum mechanics

The analysis applies wherever observations are incompatible — not only in quantum mechanics. Orthomodular structure arises in contextual behavioural science, distributed systems with interfering sensors, database queries with access restrictions, and experimental designs with mutually exclusive protocols. In each setting, the question “does empirical adequacy entail value-definiteness?” has the same algebraic answer: only if the observation algebra is distributive.

Open questions

Several directions remain:

1. *The OML extension problem.* Does every state on a general OML extend to a σ -additive measure on $S_0(A)$, as in the Boolean case? Gleason’s theorem answers this for $L(H)$; the general case is open (§6).
2. *Intermediate algebras.* Between Boolean algebras and arbitrary OMLs lie the MO_n -free OMLs, modular ortholattices, and other subvarieties. Where exactly does VDR fail in this spectrum?
3. *Dynamical structure.* If the observation algebra carries a group action (stationarity, exchangeability), how does symmetry interact with the coherence hierarchy? The Boolean case is partially understood (de Finetti’s theorem for exchangeability); the OML case is unexplored.
4. *The topos connection.* The Döring–Isham topos approach [4] addresses closely related questions via spectral presheaves rather than the McDonald–Bimbó filter-space duality. A comparison of the two frameworks — and whether the commensurability theorem has a topos-theoretic formulation — is a natural next step.

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